

Fundamentals of Battery, SOC Estimation and Battery Management Systems(BMS)

Submitted by:

Shrijan Samrat

Summer Intern,NIT-C

GUIDE: Dr. Nimal Madhu

1. Abstract:

This report presents the fundamental concepts of batteries and Battery Management Systems (BMS) with a focus on State of Charge (SOC) estimation in lithium-ion batteries. The report covers important battery parameters such as voltage, current, capacity, charging and discharging processes, SOC, State of Health (SOH), and Remaining Useful Life (RUL). It also discusses the working principles of lithium-ion batteries, different battery chemistries, and the importance of accurate SOC estimation in electric vehicles and modern battery systems. In addition, the role of BMS in battery monitoring, safety, and intelligent energy management is explained. These concepts form the foundation for machine learning-based SOC estimation and Edge-AI applications in battery systems.

2. Introduction:

Batteries have become an essential part of modern electronic systems and electric vehicles due to the increasing demand for portable and sustainable energy storage solutions. Among different battery technologies, lithium-ion batteries are widely used because of their high energy density, lightweight structure, and rechargeable characteristics. These batteries are commonly found in electric vehicles (EVs), mobile phones, laptops, renewable energy systems, and industrial applications.

Efficient battery monitoring and management are important for ensuring battery safety, performance, and long operational life. Important battery parameters such as voltage, current, capacity, State of Charge (SOC), State of Health (SOH), and Remaining Useful Life (RUL)

are continuously monitored using a Battery Management System (BMS). Accurate estimation of these parameters is critical for improving battery reliability and user confidence in electric vehicles.

However, battery behavior is complex and nonlinear because it is affected by temperature, charging and discharging conditions, battery aging, and load variations. Therefore, intelligent approaches such as machine learning and Edge-AI are increasingly being explored for accurate SOC estimation and battery monitoring applications.

This report presents the fundamental concepts of batteries, battery management systems, lithium-ion batteries, and battery chemistries with a focus on SOC estimation in modern EV systems.[1]

3. Battery Fundamentals:

3.1 Battery:

A battery is an electrochemical device that converts stored chemical energy into electrical energy. Batteries are widely used in portable electronic devices, electric vehicles, renewable energy systems, and industrial applications. A typical battery consists of four main components: anode, cathode, electrolyte, and separator. During battery discharge, electrons flow from the anode to the cathode through an external circuit and supply electrical energy to connected devices. During charging, external electrical energy is used to reverse the chemical reactions and store energy again inside the battery.

3.2 Voltage:

Voltage is the electrical potential difference between two points in a circuit and is responsible for driving electric current. It represents the electrical push that causes electrons to flow through a circuit. Voltage is measured in volts (V). In batteries, voltage is an important parameter because it reflects the electrical energy available for powering devices and is widely used for SOC estimation.

3.3 Current:

Electric current is the flow of electric charge through a circuit and is measured in amperes (A). In battery systems, current flows during both charging and discharging operations. High current flow can increase battery temperature and affect battery life and performance. Current measurement is important in Battery Management Systems for battery monitoring and SOC estimation.

3.4 Capacity:

Battery capacity represents the amount of electrical charge that a battery can store and deliver over time. It is commonly measured in ampere-hours (Ah). Batteries with higher capacity can provide power for a longer duration. Battery capacity is directly related to the State of Charge (SOC), which indicates the remaining available capacity of a battery.

3.5 Charging and Discharging:

Charging is the process of storing energy inside a battery using an external power source. During charging, the battery SOC and voltage generally increase. Discharging is the process in which the battery supplies electrical energy to an external device. During discharging, the stored chemical energy is converted into electrical energy, and the SOC gradually decreases. Charging and discharging characteristics are important for battery monitoring and SOC estimation.

4. Battery Parameters:

4.1 State of Charge (SOC):

State of Charge (SOC) represents the remaining available charge in a battery as a percentage of its total capacity. SOC is one of the most important parameters in Battery Management Systems and electric vehicles because it indicates the current energy level of the battery.

Accurate SOC estimation helps improve battery safety, driving range prediction, charging control, and user reliability.[2]

However, SOC estimation is challenging because battery behavior is affected by temperature, aging, charging conditions, and load variations.

SOC Equation:

$$\text{SOC (\%)} = (\text{Remaining Capacity} / \text{Total Capacity}) \times 100$$

4.2 State of Health (SOH):

State of Health (SOH) indicates the overall condition and aging status of a battery compared to a new battery. As batteries age, their capacity decreases and internal resistance increases, resulting in reduced performance and efficiency.[3]

SOH monitoring helps estimate battery degradation and maintain reliable battery operation.

Accurate SOH estimation is also important because old batteries behave differently from new batteries during charging and discharging operations.

4.3 Remaining Useful Life (RUL)

Remaining Useful Life (RUL) represents the estimated operational life remaining in a battery before replacement becomes necessary. RUL prediction is important for battery maintenance, safety, and reliability analysis in electric vehicles and industrial battery systems. Machine learning techniques are increasingly being explored for predicting battery degradation and estimating RUL under varying operating conditions.[2]

5. Battery Management System (BMS):

A Battery Management System (BMS) is an intelligent electronic system used to monitor, control, and protect battery operation. BMS plays a critical role in ensuring battery safety, performance, and long operational life in electric vehicles and modern battery-powered systems. It continuously monitors important battery parameters such as voltage, current, temperature, SOC, and SOH during charging and discharging operations.

5.1 Functions of BMS:

The major functions of a Battery Management System include SOC estimation, SOH monitoring, temperature monitoring, charging and discharging control, cell balancing, and battery protection. BMS protects the battery from abnormal operating conditions such as

overcharging, deep discharge, overheating, and excessive current flow. It also communicates important battery information to users and external control systems.

5.2 Functional Block Diagram of BMS:

The functional block diagram of a Battery Management System consists of battery sensors, a controller unit, estimation algorithms, protection circuits, and communication modules.

Voltage, current, and temperature sensors continuously monitor battery conditions and send data to the controller. The controller processes the sensor data to estimate SOC, SOH, and battery condition. Protection circuits ensure safe operation, while communication modules transfer battery information to users and external systems.

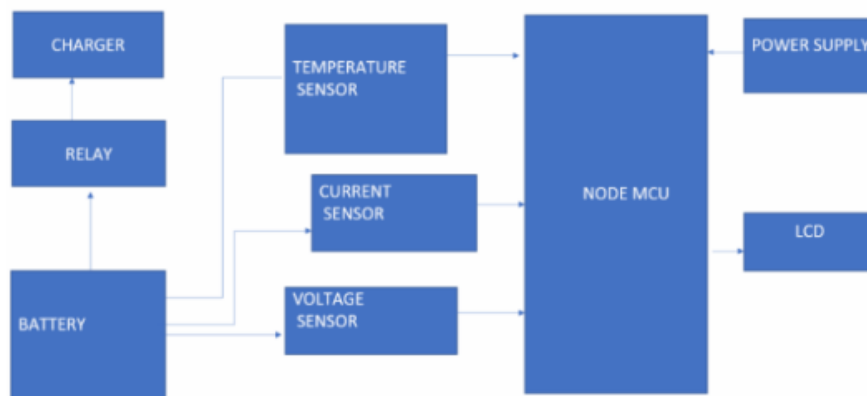


Figure 1: Functional Block Diagram of Battery Management System (BMS)

6. Lithium-Ion Battery:

Lithium-ion batteries are rechargeable batteries widely used in electric vehicles, mobile phones, laptops, renewable energy systems, and portable electronics. These batteries are popular because of their high energy density, lightweight structure, long cycle life, and efficient charging characteristics. Lithium-ion batteries operate using the movement of lithium ions between the anode and cathode during charging and discharging processes.

6.1 Working Principle:

A lithium-ion battery mainly consists of an anode, cathode, electrolyte, and separator. During charging, lithium ions move from the cathode to the anode and energy is stored inside the battery. During discharging, lithium ions move back from the anode to the cathode while electrons flow through the external circuit to power connected devices. The charging and discharging behavior of lithium-ion batteries is nonlinear and affected by temperature, battery aging, and load conditions, making accurate SOC estimation challenging.

6.2 Advantages and Limitations:

Lithium-ion batteries provide several advantages such as high energy density, lightweight construction, low self-discharge, and rechargeable operation.[1]

These characteristics make them highly suitable for electric vehicles and portable electronics. However, lithium-ion batteries are sensitive to overheating, overcharging, and deep discharge conditions. In extreme situations, excessive heat generation may lead to thermal runaway, which can cause battery damage or fire hazards.[4]

Therefore, Battery Management Systems are essential for safe lithium-ion battery operation.

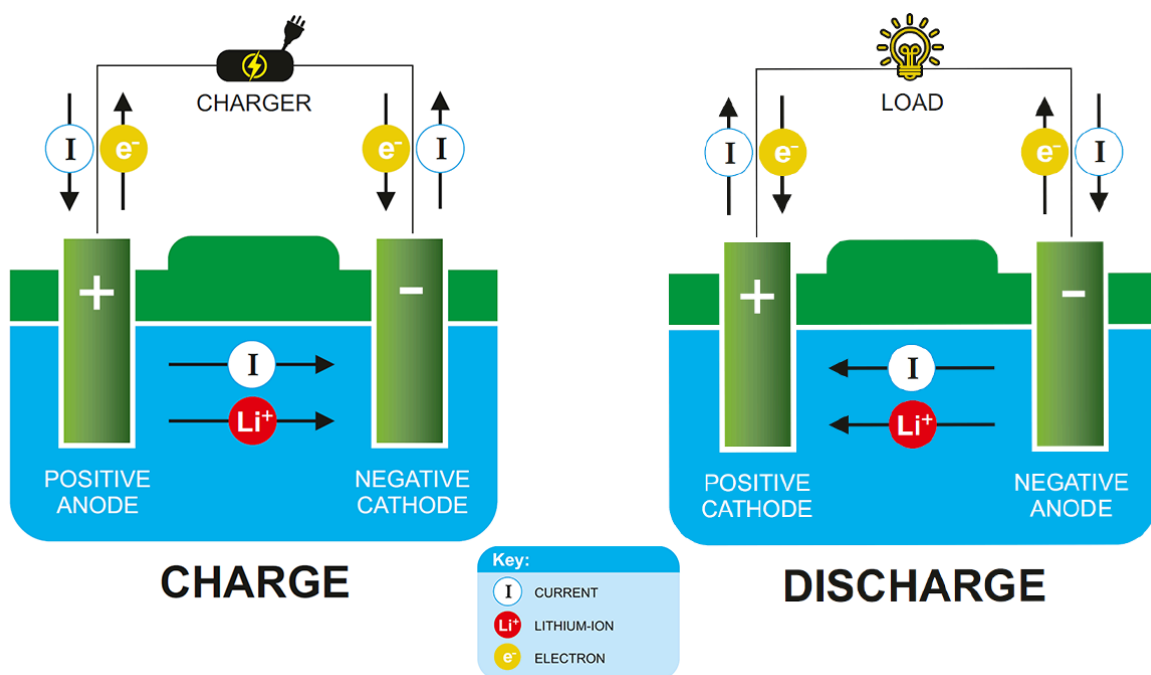


Figure 2: Basic Working Principle of Lithium-Ion Battery:

7. Battery Chemistries

7.1 Lead-Acid Battery:

Lead-acid batteries are one of the oldest rechargeable battery technologies and are commonly used in automobiles, UPS systems, and backup power applications. They are inexpensive and reliable but have low energy density and heavy weight compared to lithium-ion batteries. Due to their large size and lower efficiency, they are less suitable for modern electric vehicles.

7.2 Lithium Iron Phosphate (LFP):

Lithium Iron Phosphate (LFP) batteries are a type of lithium-ion battery chemistry known for their high safety, thermal stability, and long cycle life. LFP batteries are widely used in electric vehicles and energy storage systems because of their reliable performance and lower risk of thermal runaway [5]. However, they generally have lower energy density compared to some other lithium-ion chemistries.

7.3 Nickel Manganese Cobalt (NMC):

Nickel Manganese Cobalt (NMC) batteries are widely used in modern electric vehicles because of their high energy density and strong performance characteristics. NMC batteries can store large amounts of energy in compact sizes, making them suitable for long-range EV applications. However, they are more expensive and require careful thermal management for safe operation.

7.4 Nickel Metal Hydride (NiMH):

Nickel Metal Hydride (NiMH) batteries are rechargeable batteries previously used in hybrid vehicles and portable electronic devices. They provide better environmental performance compared to older battery chemistries and offer good safety characteristics. However, NiMH batteries have lower energy density and efficiency compared to lithium-ion batteries.

7.5 Solid-State Batteries:

Solid-state batteries are next-generation battery technologies that use solid electrolytes instead of liquid electrolytes. These batteries have the potential to provide higher energy density, improved safety, and longer battery life compared to conventional lithium-ion batteries. Although still under development, solid-state batteries are considered promising for future electric vehicle applications.

7.6 Detailed Battery Chemistry Comparison:

Different battery chemistries exhibit different electrical, thermal, and safety characteristics. Parameters such as nominal voltage, charging voltage, energy density, cycle life, and thermal stability strongly influence battery performance and SOC estimation behavior. Therefore, understanding chemistry-specific battery characteristics is important for designing accurate Battery Management Systems and intelligent SOC estimation models.[6][7]

Table 1: Comparison of Different Battery Chemistries:

Chemistry	Nominal Voltage	Charging/Cutoff Voltage	Energy Density	Cycle Life	Thermal Stability	Safety
Lead-Acid	2.0V/cell	2.4V	Low	300–500	Moderate	Good
LFP	3.2V	3.65V	Medium	2000+	Very High	Excellent
NMC	3.6–3.7V	4.2V	High	1000–2000	Moderate	Moderate
NiMH	1.2V	1.4V	Medium	500–1000	Good	Good
Solid-State	~3.7V	Varies	Very High	High	Very High	Excellent

7.7 Voltage vs SOC Characteristics :

Voltage versus State of Charge (SOC) behavior varies significantly across different battery chemistries. In Lithium Iron Phosphate (LFP) batteries, the voltage curve remains relatively flat over a large SOC range, making voltage-based SOC estimation difficult. In contrast, Nickel Manganese Cobalt (NMC) batteries exhibit a more nonlinear sloped voltage curve, allowing easier SOC estimation using voltage measurements. Lead-acid batteries show gradual voltage reduction during discharge. These chemistry-dependent voltage characteristics create major challenges in designing universal SOC estimation models.

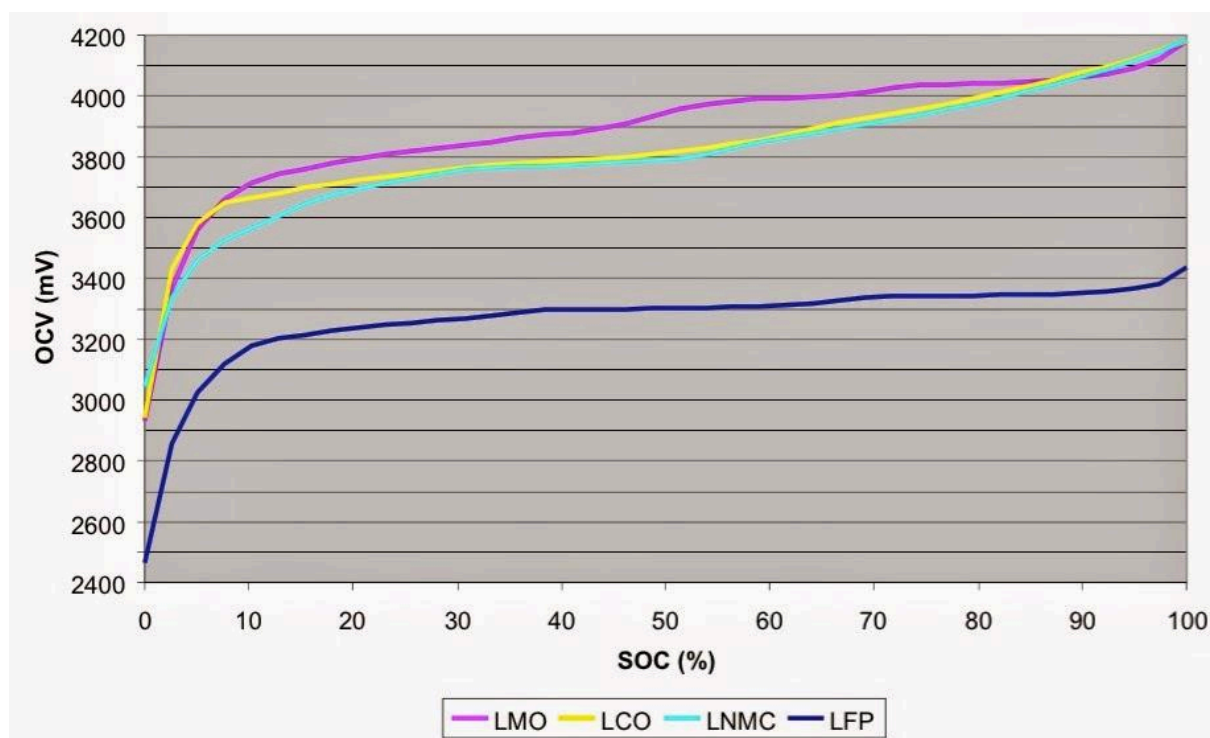


Figure 3: Comparison of Voltage vs SOC Characteristics for Different Battery Chemistries

7.8 OCV vs SOC Characteristics :

Open Circuit Voltage (OCV) is the terminal voltage of a battery measured under no-load resting conditions. OCV is strongly related to the electrochemical state of the battery and is widely used for SOC estimation in Battery Management Systems. The OCV-SOC relationship varies significantly across different battery chemistries due to differences in electrochemical reactions and voltage characteristics.

In Lithium Iron Phosphate (LFP) batteries, the OCV curve remains relatively flat over a large SOC range, making accurate voltage-based SOC estimation difficult. In contrast, Nickel Manganese Cobalt (NMC) batteries exhibit more distinguishable OCV variations with SOC. Therefore, chemistry-specific OCV modeling is important for accurate SOC estimation and battery monitoring applications.

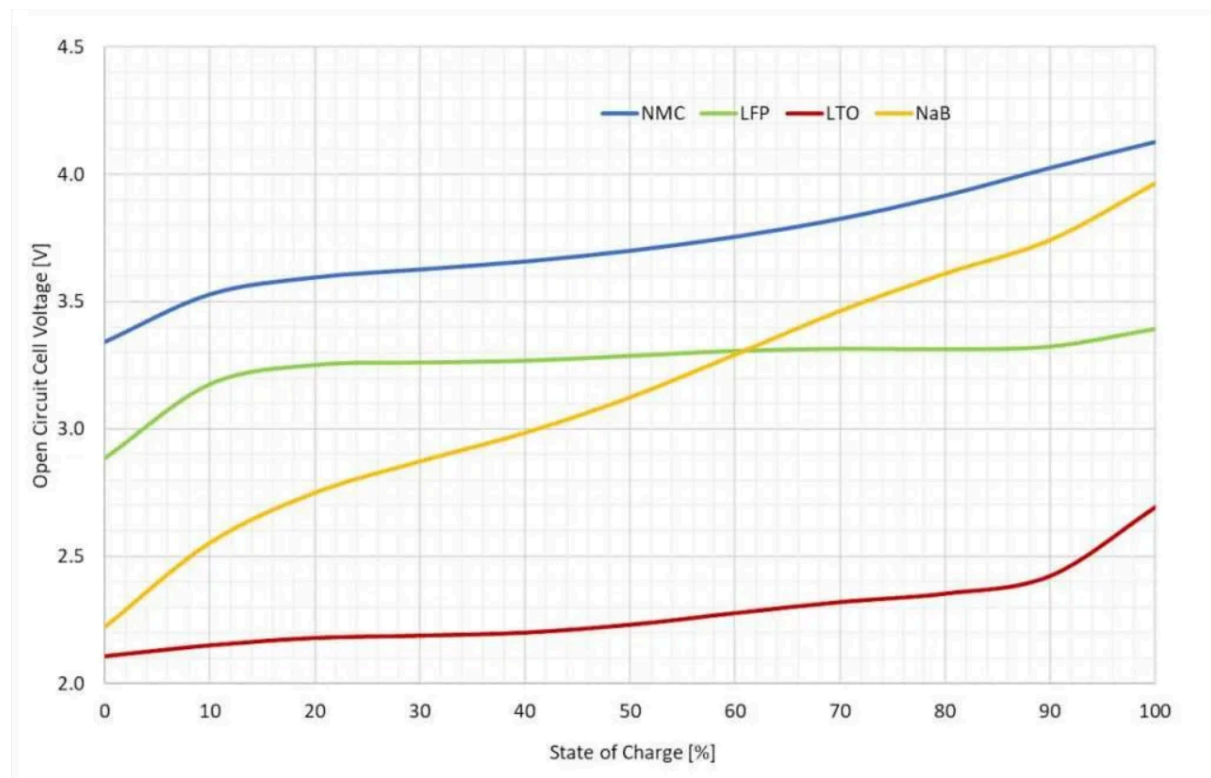


Figure 4: OCV-SOC graph

7.9 Charge and Discharge Characteristics :

Charging and discharging characteristics of lithium-ion batteries depend strongly on battery chemistry, operating temperature, load conditions, and charging rates. During charging, lithium ions move from the cathode to the anode, resulting in gradual increase in battery terminal voltage until the cutoff voltage is reached. During discharging, lithium ions move back from the anode to the cathode while the stored electrochemical energy is converted into electrical energy, causing the terminal voltage to decrease over time.

Different battery chemistries exhibit different charging and discharging behaviors. Lithium Iron Phosphate (LFP) batteries generally show flat discharge voltage plateaus over a wide SOC range, making voltage-based SOC estimation more difficult. In contrast, Nickel

Manganese Cobalt (NMC) batteries exhibit smoother nonlinear voltage reduction during discharge, allowing comparatively easier voltage-based SOC estimation.

Charging and discharging curves are important for battery modeling, thermal analysis, Battery Management System design, and SOC estimation because they reflect the electrochemical behavior and dynamic response of the battery under varying operating conditions. Accurate analysis of these characteristics is essential for improving battery safety, efficiency, cycle life, and real-time battery monitoring applications.[6]

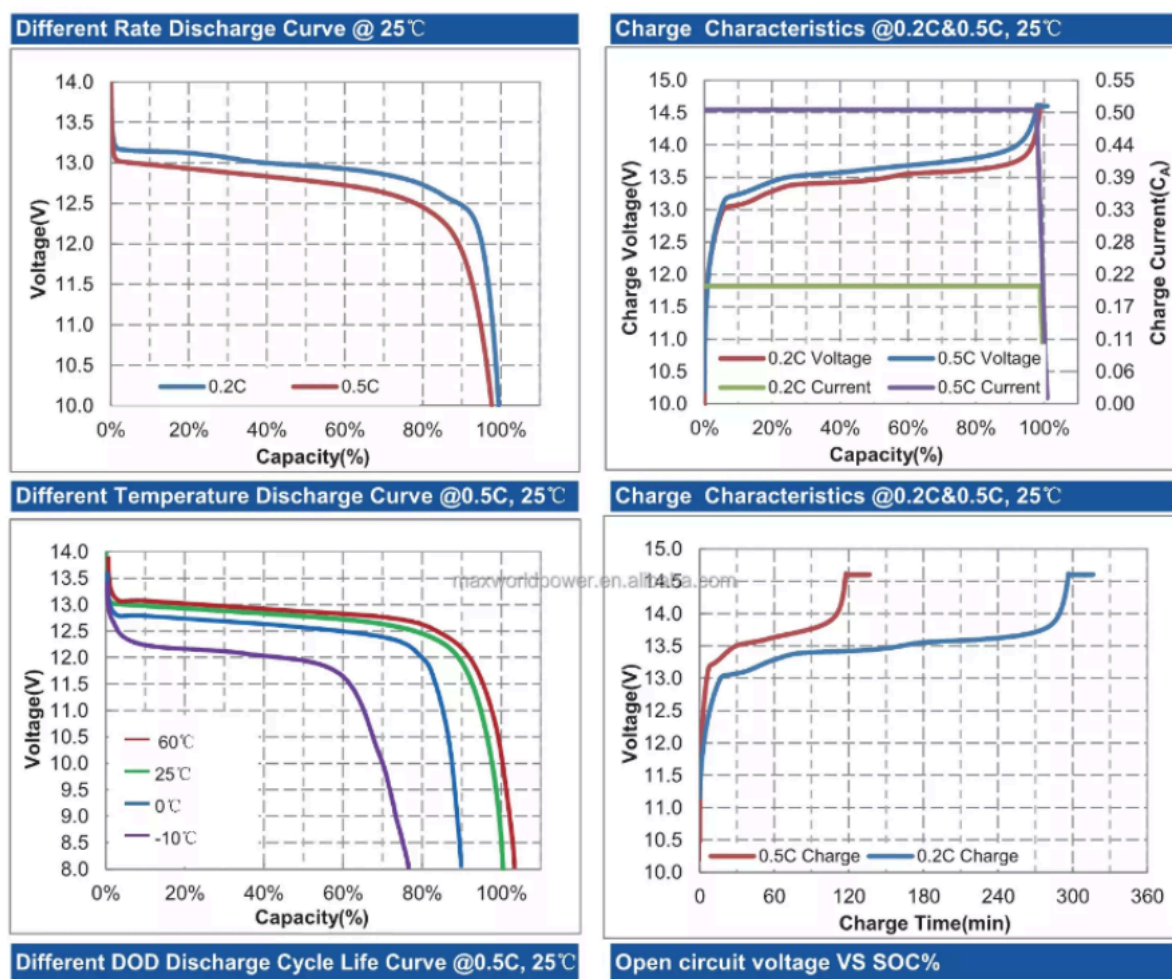


Figure 5: Charging and Discharging Characteristics of LFP and NMC Batteries

8.Thermal Behavior and Internal Resistance :

Battery thermal behavior and internal resistance significantly affect battery safety, efficiency, performance, and SOC estimation accuracy. During charging and discharging operations, electrochemical reactions inside the battery generate heat due to internal resistance and current flow. Excessive heat generation can increase battery temperature and accelerate battery degradation.

In extreme operating conditions such as overcharging, short circuits, or high-temperature environments, uncontrolled heat generation may lead to thermal runaway. Thermal runaway is a dangerous condition in which rapid temperature increase triggers self-sustaining exothermic reactions inside the battery, potentially causing fire hazards, gas release, or battery failure. Therefore, thermal monitoring is an important function of modern Battery Management Systems.

Internal resistance of a battery varies with State of Charge (SOC), operating temperature, charging/discharging current, and battery aging conditions. As batteries age, internal resistance generally increases due to electrochemical degradation and material changes inside the battery. Increased internal resistance reduces charging efficiency, increases heat generation, and affects battery voltage response during operation.

Temperature-sensitive battery behavior and resistance variation make accurate SOC estimation challenging under real-world operating conditions. Therefore, thermal management and resistance-aware battery modeling are important for developing reliable Battery Management Systems and intelligent SOC estimation methods.[3][4]

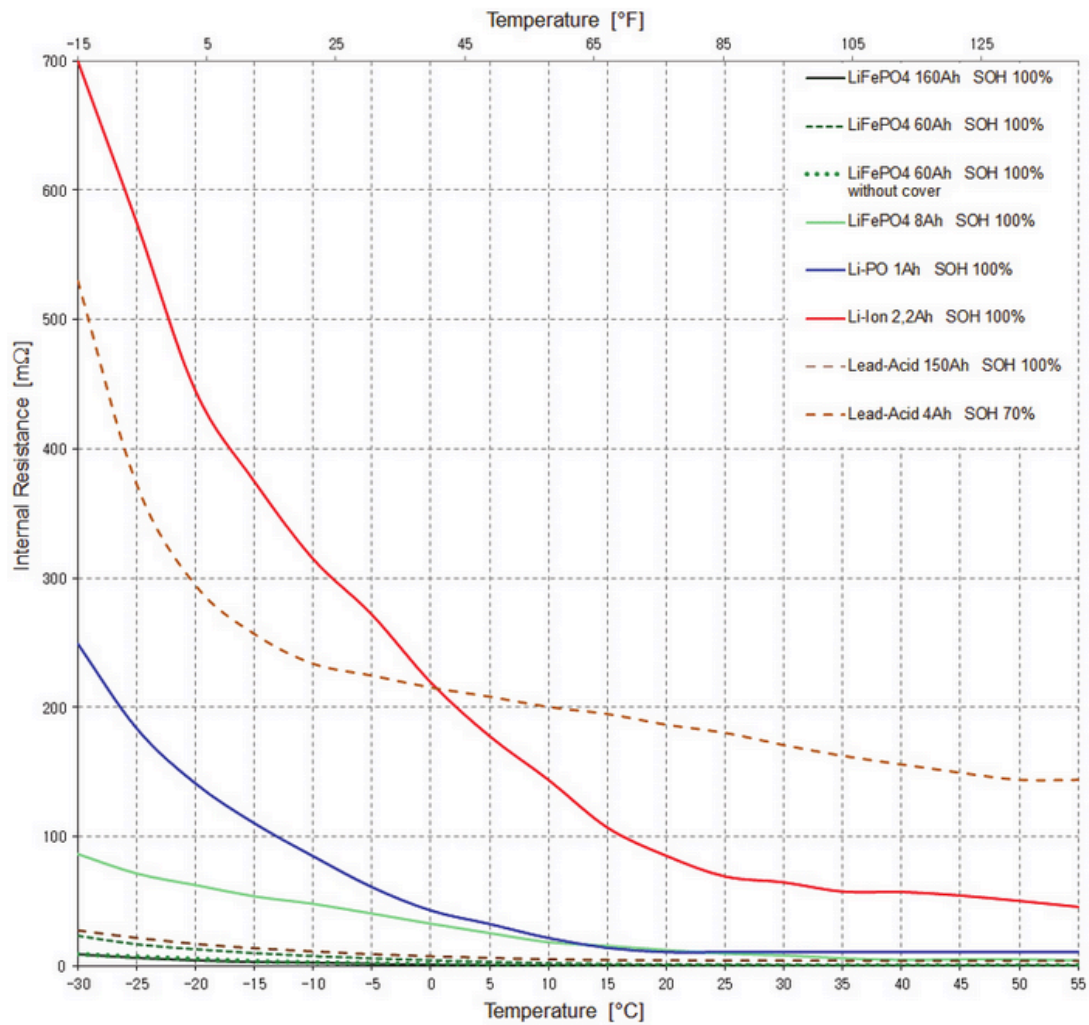


Figure 6: Variation of Internal Resistance with Temperature and Aging

9. Importance of Accurate SOC Estimation:

Accurate State of Charge (SOC) estimation is one of the most important requirements in Battery Management Systems for electric vehicles and modern battery-powered systems. SOC estimation helps determine the remaining available energy in a battery and plays a critical role in driving range prediction, charging control, battery protection, and user safety.

Incorrect SOC estimation may result in unexpected battery shutdown, reduced battery life, inefficient charging, and poor user experience. Accurate SOC

monitoring improves battery reliability and increases user confidence in electric vehicles.

However, accurate SOC estimation is difficult because battery behavior is nonlinear and influenced by several factors such as voltage, current, temperature, battery chemistry, aging, and load conditions. Traditional estimation methods may not always provide accurate results under varying operating conditions.

Therefore, machine learning and Edge-AI based approaches are increasingly being explored for intelligent SOC estimation. Machine learning models can learn complex battery behavior patterns from battery data and improve SOC prediction accuracy in real-time applications.[2]

Simple Voltage-Based SOC Estimation Equation:

$$\text{SOC} = ((\text{Voltage} - 3.0) / (4.2 - 3.0)) \times 100$$

10.Challenges in Multi-Chemistry SOC Estimation:

Accurate State of Charge (SOC) estimation across multiple battery chemistries is one of the major challenges in modern Battery Management Systems.

Different battery chemistries exhibit significantly different electrochemical behavior, voltage characteristics, thermal properties, and aging patterns. These chemistry-dependent variations make it difficult to develop a universal SOC estimation model suitable for all battery types.

Different batteries such as Lithium Iron Phosphate (LFP), Nickel Manganese Cobalt (NMC), and Lead-Acid batteries show different voltage versus SOC relationships. LFP batteries exhibit relatively flat voltage plateaus over large SOC ranges, while NMC batteries show smoother nonlinear voltage variation. As a result, SOC estimation methods based only on terminal voltage may perform differently for different chemistries.

Battery nonlinearities further increase estimation complexity. Battery voltage, internal resistance, and dynamic response vary with temperature, charging/discharging current, load conditions, and battery aging. Over time, electrochemical degradation changes battery characteristics, making SOC estimation models less accurate if aging effects are not considered.

Another important challenge is transfer learning and model generalization across different battery chemistries. Machine learning models trained using data from one chemistry may not accurately estimate SOC for another chemistry because the underlying electrochemical behavior and voltage characteristics are different. Therefore, chemistry-aware adaptive models are required for reliable multi-chemistry SOC estimation.

In addition, implementation of advanced SOC estimation algorithms on embedded Battery Management Systems introduces hardware limitations such as low memory availability, limited computational power, and real-time processing constraints. Therefore, lightweight and efficient TinyML and Edge-AI based approaches are increasingly being explored for real-time SOC estimation in modern battery systems.[6]

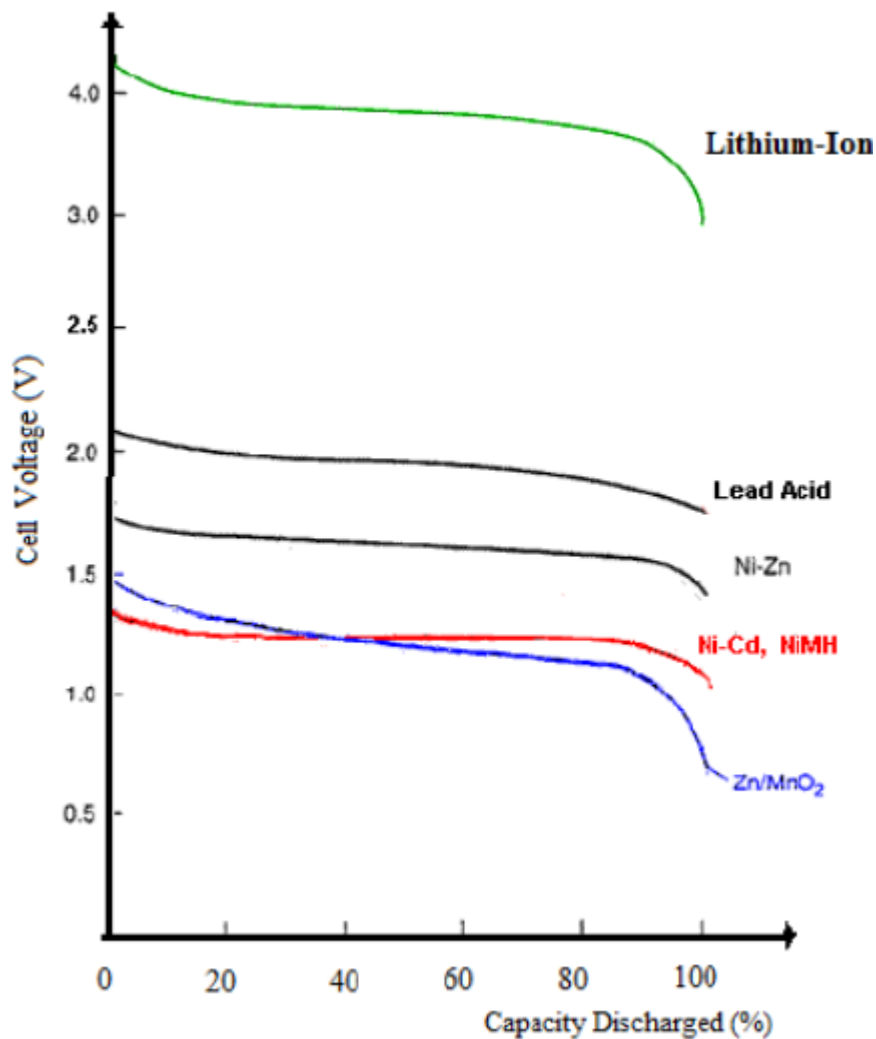


Figure 7: Comparison of Voltage-SOC Characteristics Across Different Battery Chemistries

11. Conclusion:

This report presented the fundamental concepts of batteries and Battery Management Systems with a focus on lithium-ion batteries and SOC estimation. Important battery parameters such as voltage, current, capacity, SOC, SOH, and RUL were discussed along with the working principles of lithium-ion batteries and different battery chemistries.

The report also explained the importance of accurate SOC estimation for electric vehicles and modern battery-powered systems. Battery behavior is highly nonlinear and influenced by several operating conditions such as temperature, charging/discharging characteristics, battery chemistry, internal resistance, and battery aging. Therefore, accurate SOC estimation remains a challenging problem in modern Battery Management Systems.

Different battery chemistries such as LFP and NMC exhibit different voltage-SOC characteristics, OCV behavior, thermal properties, and charge/discharge responses. These chemistry-dependent variations make universal SOC estimation difficult and require chemistry-aware battery modeling approaches.

Modern Battery Management Systems increasingly rely on intelligent estimation methods, thermal monitoring, and advanced battery analytics to improve battery safety, efficiency, and operational reliability. In addition, lightweight TinyML and Edge-AI based machine learning approaches are emerging as promising solutions for real-time SOC estimation on embedded battery systems with limited computational resources.

These concepts provide a strong technical foundation for further study and development of multi-chemistry physics-informed SOC estimation systems for future intelligent battery applications.[6]

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